

From 15,099 player-seasons to a data-driven shortlist

All filtering is automated using data within our own dataset. Player ages are calculated from birth_date in player_info.csv — no external research required to remove veterans.

1.57 pts/60

League avg efficiency threshold (2017-2019)

33.74 hrs

League avg TOI/season threshold (2017-2019)

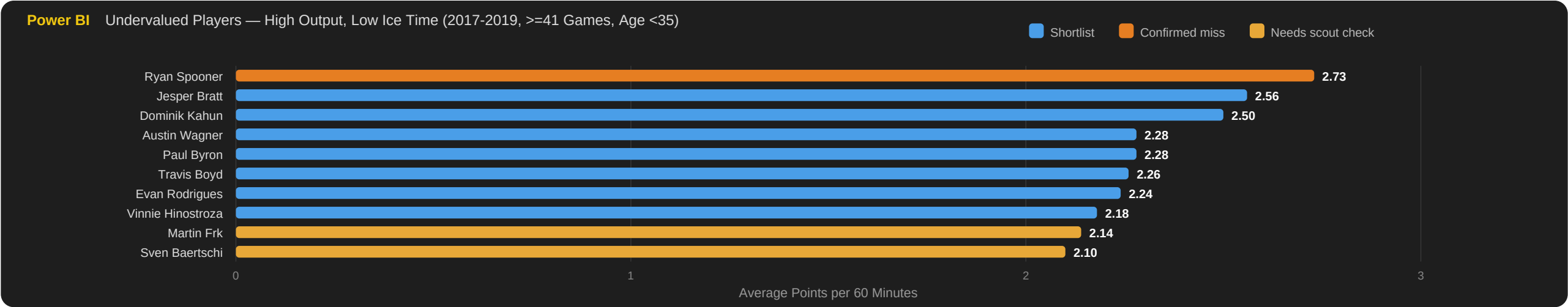
Age < 35

Veteran cutoff from birth_date in dataset

10 players

Shortlist after all automated filters

POWER BI DASHBOARD · mart_player_season · pts/60 > 1.57 avg · TOI < 33.74 hrs avg · >=41 games · 2017-2019 · is_veteran = False (age < 35)



How is_underused_high_impact is calculated

pts/60 > 1.57 (league avg) AND total season TOI < 33.74 hrs (league avg). Both thresholds recalculate each season dynamically.

How is_veteran is calculated

Age from birth_date in player_info.csv: FLOOR(DATEDIFF(days, birth_date, 2019-12-31) / 365.25) >= 35. Removes veterans automatically — no external research needed.

10 names on the shortlist — what happened after 2019?

All filters automated from within the dataset. Ages from player_info.csv remove veterans (age >=35). Dataset ends 2019 — here is what happened next.

REAL-WORLD OUTCOMES AFTER 2019 · Verified June 2026

Jesper Bratt Age 21 at end of dataset New Jersey Devils franchise cornerstone & alternate captain. 168G/336A in 622 games. 2.56 pts/60 Franchise cornerstone	Evan Rodrigues Age 26 at end of dataset Won the Stanley Cup with Colorado Avalanche 2022. Currently Florida Panthers. 2.24 pts/60 Stanley Cup winner 2022
Travis Boyd Age 26 at end of dataset Career-high 17G/35pts with Arizona — double the minutes Washington gave him. 2.26 pts/60 Breakout elsewhere	Paul Byron Age 30 at end of dataset Named alternate captain at Montreal Canadiens. 98 career goals, 208 career points. 2.28 pts/60 Alternate captain
Vinnie Hinostroza Age 25 at end of dataset Still active NHL 2025-26 with Florida Panthers. Double-digit goals twice. 2.18 pts/60 Still active NHL	Dominik Kahun Age 24 at end of dataset Contributed for Pittsburgh & Edmonton. 83 career NHL points. Now Lausanne HC. 2.50 pts/60 Multi-team contributor
Austin Wagner Age 22 at end of dataset Reliable bottom-6 forward. Now playing in the KHL (Shanghai Dragons). 2.28 pts/60 Bottom-6 career	Ryan Spooner Age 27 at end of dataset Left NHL after 2019. Now in the KHL. Never sustained performance across teams. 2.73 pts/60 Confirmed miss
Martin Frk Age 26 at end of dataset 9:57 avg ice time — coaches limited due to defensive deficiencies (career -26). Now AHL. 2.14 pts/60 Defensive limitations	Sven Baertschi Age 25 at end of dataset 13th overall pick. 18G/35pts in 68 games when healthy (2016-17). 0.5 pts/game healthy rate. Recurring concussions ended career at 30. 2.10 pts/60 Talent derailed by injury

7 of 10 shortlisted players had meaningful or notable NHL careers after 2019

Filtering fully automated from data within the dataset. Evan Rodrigues won the Stanley Cup in 2022. Travis Boyd doubled his output when given real ice time. Sven Baertschi produced 0.5 pts/game when healthy — the pipeline found genuine talent whose career was derailed by injury, not a miss.

Where do shots actually become goals?

The slot converts at 19.46% — six times higher than wing shots. Practice time should follow the data, not tradition.

19.46%

Slot conversion rate

6x

More effective than the wings

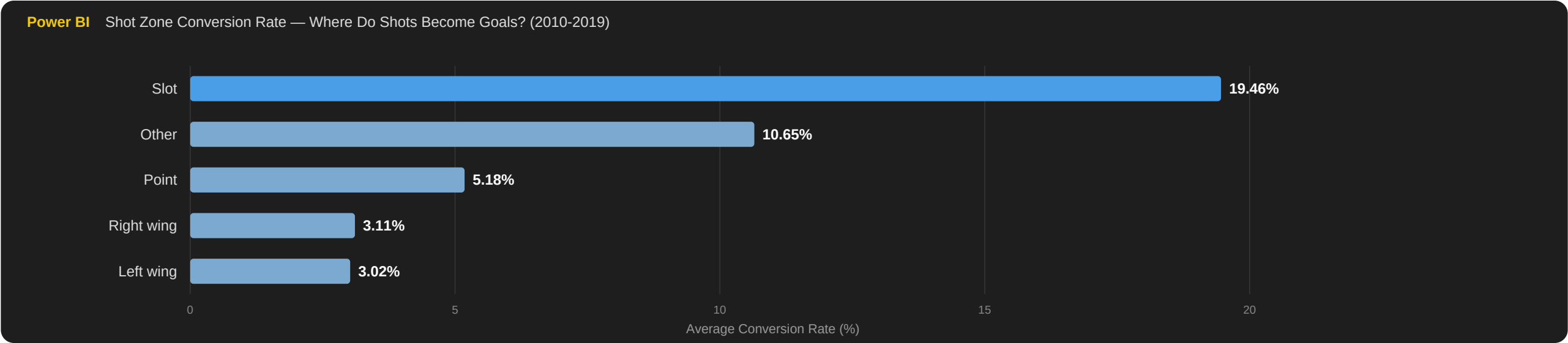
9 vs 33

Slot vs wing shots per goal

1.7M+

Shot events analysed (2010-19)

POWER BI DASHBOARD · mart_shot_zones · conversion rate by zone, 2010-2019



The finding

Slot shots convert at 19.46% vs ~3% from the wings. The hierarchy holds across every season 2010-2019. Pre-2010 excluded (100% missing coordinates).

The recommendation

Drill slot positioning and finishing, not perimeter technique. The problem is rarely the shot — it is where the player is standing when they take it.

When do penalties actually cost games?

A Period 3 Game Misconduct correlates with losing two times out of three. A Period 1 Minor is a coin flip. Severity and timing both matter.

68.65%

Period 3 Game Misconduct loss rate

50.82%

Period 1 Minor — a coin flip

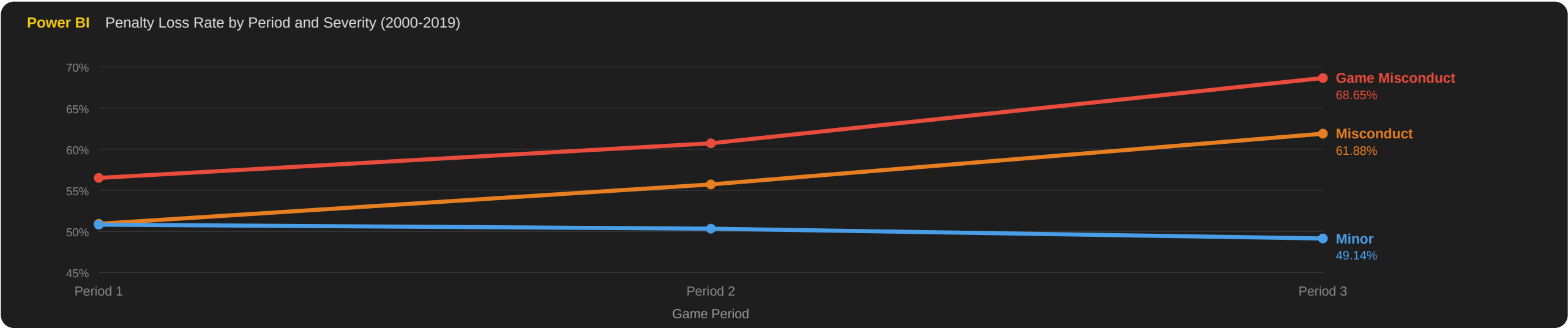
12pp

P3 vs P1 gap, same penalty type

247K+

Penalty records analysed

POWER BI DASHBOARD · mart_penalty_cost · loss rate by period and severity, 2000-2019



Correlation, not causation

Teams already losing may take more desperate penalties. But the same penalty type carries a 12-point higher loss rate in Period 3 than Period 1 — if score state alone explained it, that gap would be far smaller.

The recommendation

Coach discipline differently by period. A Period 1 Minor is noise. A Period 3 Game Misconduct costs the game two times out of three. The pre-Period-3 locker room message should be specifically about composure.

How much does home ice actually matter?

Home ice advantage is worth +13 to +22 percentage points in win probability at top NHL venues — and nearly a full extra goal per game.

+13-22pp

Win rate uplift at top home venues

+0.9

More goals per game at home

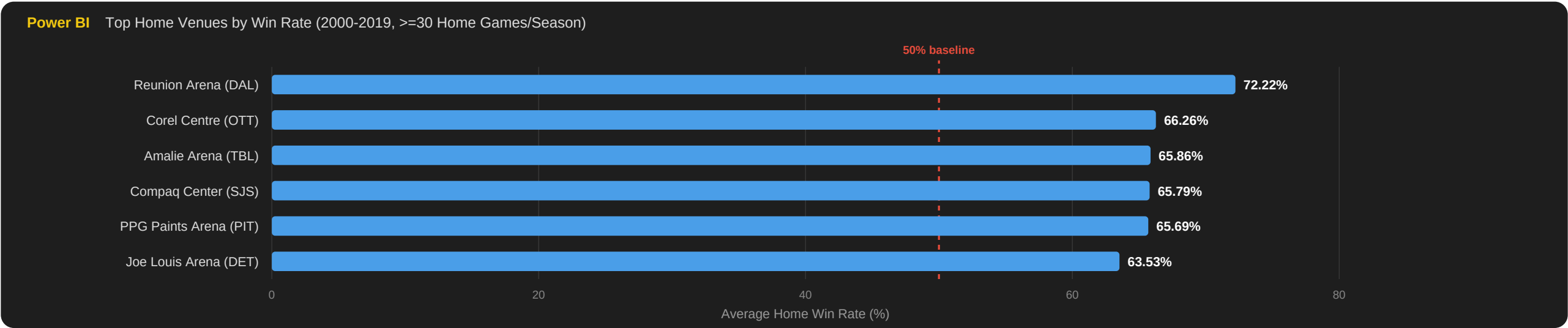
65.86%

Tampa Bay Lightning at Amalie Arena

All teams

Every venue between 55-72% home wins

POWER BI DASHBOARD · mart_venue_advtg · top home venues by win rate, >=30 home games/season, 2000-2019



The finding

Every qualifying venue wins between 55% and 72% at home — home advantage is league-wide, not arena-specific. Home teams score ~0.9 more goals per game. (Reunion Arena is a small-sample outlier.)

The recommendation

Finishing top of the division to earn home playoff games is worth more than most trade deadline acquisitions. In a Game 7, home ice is the difference between a coin flip and winning two times out of three.

Which franchises are rising or falling?

A 3-year rolling win rate classifies every franchise by consistent directional momentum as of 2019 — before the standings confirmed it.

9 / 12 / 12

Rising / stable / falling franchises

+11.17pp

Colorado Avalanche
— largest rise

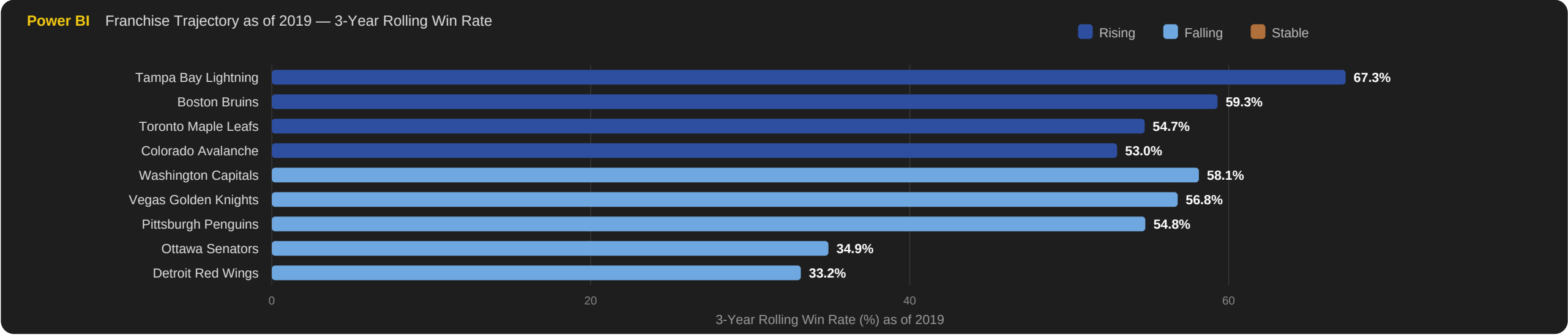
-10.07pp

Ottawa Senators
— steepest fall

momentum

Consistent 3-season direction, not net change

POWER BI DASHBOARD · mart_team_traject_rolling · 3-year rolling win rate as of 2019, colour-coded by trajectory



Trajectory measures consistent momentum, not net change
A small but unbroken decline counts as falling; a larger but inconsistent swing counts as stable. This page shows the top rising and falling franchises as of 2019. See page 2 for what happened next.

Our dataset ends in 2019 — what happened next?

The pipeline classified franchises in 2019. We checked the real-world outcomes through 2026. A compass, not a crystal ball.

RISING FRANCHISES — VERIFIED OUTCOMES AFTER 2019

Tampa Bay Lightning Won back-to-back Stanley Cups in 2020 and 2021. Champion +9.16pp	Colorado Avalanche Won the Stanley Cup in 2022 — largest upward momentum of any team. Champion +11.17pp	Boston Bruins Set the all-time NHL regular season wins record in 2022-23 (65 wins). Record-setter +5.30pp
Carolina Hurricanes Seven consecutive playoff appearances through 2025. Contender +7.97pp	Toronto Maple Leafs Eight straight playoffs; first series win since 2004 in 2023. Contender +7.56pp	New York Islanders Back-to-back Conference Finals in 2020 and 2021. Contender +2.30pp

FALLING FRANCHISES — VERIFIED OUTCOMES AFTER 2019

Detroit Red Wings 10 consecutive seasons without playoffs — longest active drought. Validated -8.53pp	Washington Capitals Have not won since 2018; now in rebuild phase. Validated -6.07pp	Pittsburgh Penguins Have not won since 2017; entering rebuild. Validated -4.73pp
Ottawa Senators Years rebuilding; returned to playoffs only in 2024-25. Validated -10.07pp	Vegas Golden Knights Won the Stanley Cup in 2023 — fell from expansion peak then rebuilt. Miss -5.90pp	New York Rangers Rebounded to reach the Conference Finals by 2024. Miss -7.07pp

In 2019, this pipeline already knew who would lift the Cup

Tampa Bay Lightning and Colorado Avalanche — both flagged rising — won within three years. Two misses (Vegas Golden Knights, New York Rangers) show the metric identifies trends, not certainties. The pipeline is a compass, not a crystal ball.